

Suraj Patel

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Portfolio: surajpatel.dev

MCA student focused on building AI-driven backend systems and real-time applications. Experienced in designing agentic workflows using LangChain and LangGraph, with hands-on work in scalable architectures, streaming systems, and production deployments on GCP.

EDUCATION

Master of Computer Applications (MCA) | Amity University Online, Noida
Bachelor of Science, | Lucknow Christian College, Lucknow

2024 – Present
2021 – 2024

PROFESSIONAL EXPERIENCE

Full Stack Engineer | Careerboat.ai

June 2025 – Present

- Developed AI interview APIs using Node.js with LangChain and LangGraph, generating context-aware questions based on resumes, skills, and prior responses, while optimizing token usage through state management using MongoDB checkpoints and dynamic context summarization.
- Integrated Google Cloud TTS, AWS S3, and SSE for real-time audio streaming and token-by-token AI responses.
- Developed an AI-powered resume analyzer to evaluate job description match rates, identify skill gaps, and dynamically generate tailored resumes.
- Integrated the Razorpay payment gateway to process secure user transactions, implementing custom backend logic to support discount coupons.

PROJECTS

AI Interview | [Source Code](#)

May 2025 - Aug 2025

- Built a full-stack AI-powered interview system using React, Node.js, and MongoDB, generating context-aware questions from resumes, skills, and user responses using LangChain and Gemini.
- Engineered a real-time interview system with WebRTC and Socket.io, integrating streaming audio and TTS, reducing next-question API response latency from ~8–10s to ~3–6s using Redis caching.

AI Manim Video Generator | [Source Code](#)

Jul 2025 - Aug 2025

- Built an end-to-end AI pipeline using React, FastAPI, and MongoDB to generate animations from text prompts into Manim code and rendered video output.
- Designed a multi-agent LangGraph workflow with validation and repair loops, enabling self-healing code generation and improving execution success from ~60% to ~90%+.
- Implemented Celery + Redis task queue for concurrent rendering, improving stability under high load, with Supabase for video storage, deployed on GCP using Docker.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python
- **Frontend:** React, Redux, HTML5, CSS3
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT, OAuth 2.0
- **Databases & Caching:** PostgreSQL (Prisma), MongoDB (Mongoose, Beanie), Redis
- **AI/ML:** LangChain, LangGraph, LangSmith, Scikit-learn, PyTorch, Pandas
- **Tools & DevOps:** Docker, Git, GitHub Actions, Celery (Task Queue System)

CERTIFICATIONS

Scrimba The Fullstack Developer Path: 2025 | [Certificate Link](#)
Scrimba React | [Certificate Link](#)